



NYU

TANDON SCHOOL
OF ENGINEERING

ME - UY 4214: Finite Element Modeling, Design, and Analysis

6:00 PM - 7:50 PM Th

6 Metrotech Room 473

Semester	Spring 2026
Class mode	In-person
Instructor	Dr. Nikhil Gupta Professor
Office hours	Thursdays, 2.00 – 3:00 PM
Contact information	Room # 132 at 5MTC, Mechanical & Aerospace Engineering Department Phone: 646-997 3080 Email: ngupta@nyu.edu

COURSE DESCRIPTION

This course introduces the **finite element method (FEM)** for modeling and solving engineering problems. Students will learn the theoretical foundations of FEM and apply them to analyze structural, thermal, and dynamic systems.

The course integrates **analytical formulation, numerical implementation, and simulation tools** to provide a comprehensive understanding of finite element modeling and analysis in mechanical engineering.

LECTURE CLASS SYLLABUS

Topics	• Introduction to Finite Element Analysis (FEA) • FEM steps and spring element • Matrix algebra and spring element examples • 2D problems and bar elements • Beam element • Energy method • 2D triangle element • Practical modeling techniques • Machine Learning in FEM
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Objectives	• Introduce the fundamental concepts of the finite element method. • Develop the mathematical formulation of finite element models. • Apply FEM to structural, thermal engineering problems.
Outcomes	Upon successful completion of this course, students will be able to: • Understand the general steps to develop the model using finite element method, • Apply the general steps to set up the solution method for linear problems using principal elements, • Apply the energy method to develop the finite element model, • Be able to write a program using the finite element method.
Textbook (optional)	<i>A First Course in Finite Element Method</i> (5 th edition or newer) by Daryl L. Logan,

LAB SYLLABUS (Prof. Dung D. Luong)

Schedule & Topics	• Week 1: Introduction to Ansys Workbench • Week 2: Mechanical Basics • Week 3: Meshing (Part I) • Week 4: Meshing (Part II) • Week 5: Contact and Connection (Part I) • Week 6: Contact and Connection (Part II) • Week 7: Boundary conditions • Week 8: Result analysis • Week 9: Thermal analysis • Week 10: Vibration analysis • Week 11: Special topic • Week 12 – 13: Lab project • Week 14: Project Presentation
Objectives	Learn the modelling procedures implemented in ANSYS to solve problems from various disciplines of mechanical engineering.
Outcomes	Students will able to: • Implement the principal elements using the ANSYS Workbench; • Construct the finite element model and analyze the complex problem in mechanical structure and heat transfer using ANSYS; • Learn the impact of FEM on engineering domain to solve real-world issues; • Be able use engineering judgment to draw conclusions.
Textbook	Not required
Software	ANSYS Workbench 2025 R2 This software is available in the NYU computer laboratories.

Installing ANSYS on personal computers is optional, although a **free student version** is available: <https://www.ansys.com/academic/students/ansys-student>

ASSIGNMENTS AND GRADING

Lecture Class	Mid-term exams (4 given, best 3 counted)	30%
	Final exam	20%
Lab	4 lab assignments	15%
	Lab project	30%
Attendance		5%
Total		100%

Course grading will be based on the tentative scale below:

100-93 A, 92-89 A-, 88-85 B+, 84-81 B, 80-77 B-, 76-73 C+, 72-70 C, 69-67 C-, 66-64 D+, 63-60 D, 59-0 F

COURSE COMMUNICATION

All course materials and announcements will be posted on **NYU Brightspace**, including:

- Lecture slides
- Assignment instructions
- Exam schedules
- Important course updates

Lecture materials will be available before class, and supporting documents will be uploaded after the lecture.

ATTENDANCE POLICY

Students who attend **at least 80% of class meetings** (>22 sessions) will receive the **5% attendance credit**.

EXAMS

- Four midterm exams will be given during the semester.
- All exams are **closed-book**; however, **one page of notes** is permitted.
- **Only the three highest scores of midterms will be counted toward the final grade.**

Exams may include:

- Multiple-choice questions
 - Analytical problem solving
 - A combination of both formats
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ASSIGNMENT

- Students can earn up to **5% extra credit** by completing **four bonus assignments**.
 - Lab assignments must be submitted through **NYU Brightspace**. **Late submissions will not receive credit.**
 - Students are responsible for completing all coursework before the end of the semester.
 - Requests for additional work after the final exam to improve a grade will not be granted.
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ACADEMIC INTEGRITY

Academic honesty is essential in engineering education. Any form of cheating, plagiarism, or academic misconduct on exams, assignments, or projects may result in **a failing grade for the course**.

Students should review the [NYU Tandon Student Code of Conduct](#) for detailed policies.

ACCESSIBILITY AND STUDENT SUPPORT

Students requesting academic accommodations should contact:

Moses Center for Student Accessibility (CSA) at mosescsa@nyu.edu

For more information: <https://www.nyu.edu/students/communities-and-groups/student-accessibility.html>

STUDENT WELLNESS AND ADVOCACY

Personal, medical, or mental health challenges can affect academic performance.

Students are encouraged to contact the: **NYU Tandon Office of Student Advocacy**

at advocacy.tandonstudentlife@nyu.edu or visit the office at **5 MetroTech Center, LC240**.
